

A Unified Perturbative Stability Classification and the Three-Tier Rigidity Hierarchy of Black Hole Structures

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Abstract

The stability of fundamental spacetime structures—event horizons, innermost stable circular orbits (ISCOs), and photon spheres—under modified gravity serves as a stringent test of a theory's physical self-consistency. We introduce a unified perturbative framework, the CH Series, to evaluate the response of these structures to specific parametric metric deformations. We rigorously demonstrate that, under a specific Type-A metric deformation on a Schwarzschild background ($g_{tt} \propto (1 - 2M/r)(1 + \alpha M/r)$), these three structures exhibit distinctly different stability levels: the event horizon is invariant at the zeroth order (ϵ^0), the first-order correction of the ISCO strictly vanishes (ϵ^2), and the first-order correction of the

photon sphere is non-vanishing (ϵ^1). We identify the mathematical root of the ISCO's first-order rigidity as the exact vanishing of the second derivative of the deformation function at the GR ISCO radius, $A_1''(6M) = 0$.

Conversely, under perturbative expansions in the spin parameter on a Kerr background, the photon sphere exhibits a first-order (a^1) response, while both the ISCO and the event horizon exhibit second-order (a^2) rigidity. This refined hierarchy provides a robust, mathematically grounded tool for testing structural stability across modified gravity theories.

Keywords: Gravitational Structural Stability; Perturbation Analysis; Innermost Stable Circular Orbit (ISCO); Photon Sphere; Event Horizon; CH Series

I. Introduction

Event horizons, photon spheres, and innermost stable circular orbits (ISCOs) serve as critical interfaces between theoretical gravity and astrophysical observations. The horizon defines the causal boundary of a black hole, the photon sphere dictates the geometric silhouette of the black hole shadow, and the ISCO determines the inner edge of the accretion disk around compact objects, alongside the red wing extension of the Fe $K\alpha$ line profile. While these structures are precisely determined in General Relativity (GR), their fate in modified gravity theories remains a central issue. Specifically: do these structures persist under slight metric deformations? Are their positional shifts first-order or higher-order effects with respect to the

perturbation parameter?

Addressing these questions is essential for the viability of any modified theory. The absence of an ISCO would invalidate a theory's ability to model compact accretion flows. High sensitivity of the photon sphere to metric corrections would render black hole shadow observations as powerful probes for modified gravity. Degradation of the event horizon under perturbations would imply that the black hole solutions in that theory are unphysical.

To address this, we develop the CH Series, a generalized framework for testing gravitational structural stability via perturbation analysis. The methodology involves applying a controllable perturbation parameter to the metric, constructing the effective potential, solving the conditions for circular orbits and stability, and subsequently expanding the critical radius as a power series of the perturbation parameter.

Under the CH Series, we have completed core tasks covering two background metrics (Schwarzschild and Kerr), three core structures, and two types of perturbations. A central finding of this work is that, under a specific parametric metric deformation on a Schwarzschild background, these three structures exhibit a distinct three-tier rigidity hierarchy. We provide rigorous analytical proofs and numerical verifications for these response orders, tracing the physical cancellations back to specific mathematical identities of the effective potential.

II. Methodology: The CH Series Perturbation Framework

2.1 Generalized Testing Protocol

The CH Series employs a six-step testing protocol applicable to any gravitational structure and perturbation type:

Step 1: Formulation of the Characteristic Equation. Construct the mathematical equation based on the physical definition of the target structure (e.g., $g_{tt} = 0$ for the horizon, $d(b^2)/dr = 0$ for the photon sphere, and $d^2V_{\text{eff}}/dr^2 = 0$ for the ISCO).

Step 2: Solving the Critical Condition. Solve the characteristic equation under the unperturbed background metric to obtain the baseline position r_0 .

Step 3: Perturbation Expansion. Apply a perturbation parameter δ to the metric, and perform a Taylor expansion of the characteristic equation around r_0 . Let $r(\delta) = r_0 + c_1\delta + c_2\delta^2 + \dots$, substitute it into the characteristic equation, and solve for the coefficients order by order.

Step 4: Evaluation of the First-Order Coefficient. Calculate the first-order coefficient c_1 .

Step 5: Second-Order and Higher-Order Analysis. If $c_1 = 0$, calculate the second-order coefficient c_2 to determine the dominant order of the structural shift.

Step 6: Stability Classification. Based on the value of c_1 , we classify the structural response as:

- **First-order Rigid (STRUCTURE-PRESERVED):** $c_1 = 0$. The structure's

position is insensitive to the perturbation at the leading order.

- **First-order Sensitive** (STRUCTURE-DEFORMED): $c_1 \neq 0$. The structure undergoes a linear shift proportional to the perturbation parameter.
- **Structural Collapse** (STRUCTURE-CRITICAL): The characteristic equation admits no real, physical solutions for r beyond a critical perturbation threshold.

2.2 Definition of Perturbation Types

The CH Series supports standard parametric perturbations:

Type A: Pure g_{tt} Deformation. Modifies the time component of the metric:

$g_{tt} = -(1 - 2M/r)(1 + \alpha M/r)$, where α is the deformation coefficient.

Type B: Joint g_{tt} and g_{rr} Deformation. Simultaneously modifies the time and radial components: $g_{tt} = -(1 - 2M/r)(1 + \alpha M/r)$ and $g_{rr} = (1 - 2M/r)^{-1}(1 + \beta M/r)^{-1}$.

III. Core Findings: The Three-Tier Rigidity Hierarchy on a Schwarzschild Background

3.1 ISCO Stability Analysis and Rigorous Proof (CH-0001)

On a Schwarzschild background, applying a Type A perturbation to g_{tt} . The ISCO condition requires both $V_{\text{eff}}'(r) = 0$ and $V_{\text{eff}}''(r) = 0$. Substituting the specific angular momentum $L^2(r, \alpha)$ into the marginal stability condition yields a characteristic equation $F(r, \alpha) = 0$.

To determine the first-order shift c_1 , we apply the Implicit Function Theorem.

Expanding $F(r, \alpha) = 0$ around the GR limit ($r_0 = 6M, \alpha_0 = 0$) yields:

$$c_1 = - \left. \frac{\partial F / \partial \alpha}{\partial F / \partial r} \right|_{r=6M, \alpha=0}$$

Core Observation: The vanishing of the first-order coefficient c_1 is not a mere algebraic accident, but a direct consequence of a specific mathematical property of the Type-A deformation. By decomposing the metric function as $A(r, \alpha) = A_0(r) + \alpha A_1(r)$, where $A_1(r) = \frac{M}{r} - \frac{2M^2}{r^2}$, we find that its second derivative evaluates to exactly zero at the GR ISCO radius:

$$A_1''(r) = \frac{2M}{r^3} - \frac{12M^2}{r^4} \implies A_1''(6M) = 0$$

This mathematical identity is the root cause of the cancellation. In the characteristic equation $F(r, \alpha) = 0$, the term governing the first-order shift is proportional to $A_1''(6M)$. Because this term vanishes, the first-order deformation of the gravitational potential is exactly counterbalanced by the first-order adjustment of the centrifugal barrier. Consequently, $c_1 = 0$ **strictly**. The second-order coefficient evaluates to $c_2 \approx 4.44$, yielding:

$$r_{\text{ISCO}}(\alpha) = 6M + 4.44M\alpha^2 + \mathcal{O}(\alpha^3)$$

Classification: First-order Rigid (STRUCTURE-PRESERVED). (*A complete step-by-step algebraic proof is provided in Appendix A*).

Numerical Verification of the Second-Order Law:

To corroborate the analytical result $c_1 = 0$, we numerically solve the exact characteristic equation $F(r, \alpha) = 0$ for small values of α . The positional shift $\Delta r_{\text{ISCO}} = r_{\text{ISCO}}(\alpha) - 6M$ is computed:

Perturbation α	Numerical $\Delta r_{\text{ISCO}}/M$	Analytical $4.44\alpha^2$	Ratio $\frac{\Delta r(\alpha)}{\alpha^2}$
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The numerical results confirm that $\Delta r_{\text{ISCO}} \propto \alpha^2$. Doubling α from 0.01 to 0.02 increases the shift by a factor of exactly ≈ 4 ($0.0017765/0.0004441 \approx 4.00$), rigorously validating the absence of any first-order (α^1) contribution.

3.2 Photon Sphere Stability Analysis (CH-S-0002)

The characteristic equation for the photon sphere is $A'(r) = 2A(r)/r$, where $A(r) = (1 - 2M/r)(1 + \alpha M/r)$. This equation possesses an exact closed-form solution. The perturbation expansion is:

$$r_{\text{ph}}(\alpha) = 3M - \frac{M\alpha}{6} + \mathcal{O}(\alpha^2)$$

The first-order coefficient $c_1 = -1/6 \neq 0$. The photon sphere exhibits first-order sensitivity to metric deformations.

Classification: First-order Sensitive (STRUCTURE-DEFORMED).

3.3 Event Horizon Stability Analysis (CH-E-0003)

The characteristic equation for the event horizon is $g_{tt} = 0$, which under Type A deformation yields:

$$\left(1 - \frac{2M}{r}\right) \left(1 + \alpha \frac{M}{r}\right) = 0$$

This equation possesses two roots: $r_1 = 2M$ and $r_2 = -\alpha M$.

To assess the physical validity of r_2 , we must consider the domain of validity of the perturbative framework. The perturbation theory assumes $\alpha \ll 1$ and is constructed to describe deviations in the physical strong-field

region $r \sim \mathcal{O}(M)$. For $\alpha > 0$, $r_2 < 0$, which lies outside the physical manifold. For $\alpha < 0$, $r_2 = |\alpha|M \ll M$. In this regime, the perturbative expansion parameter $\alpha M/r$ becomes $\mathcal{O}(1)$, violating the fundamental assumption of perturbation theory ($\alpha M/r \ll 1$). Therefore, r_2 is an artifact of perturbative extrapolation beyond its radius of convergence, not a genuine physical horizon. The physical event horizon remains strictly at $r = 2M$, exhibiting a **zeroth-order response**.

Classification: Zeroth-order Rigid (STRUCTURE-PRESERVED).

3.4 Summary of the Schwarzschild Hierarchy

Structure	Response Order	First-order Coefficient	Stability Classification
Event Horizon	ϵ^0 (Zeroth-order invariant)	0	First-order Rigid
ISCO	ϵ^2 (First-order cancellation)	0	First-order Rigid
Photon Sphere	ϵ^1 (First-order non-vanishing)	-1/6	First-order Sensitive

IV. Spin Perturbation Analysis on a Kerr Background

4.1 ISCO and Horizon Response to Spin (CH-S-0003 & CH-E-0004)

In the Kerr background, we perform a perturbative expansion in the dimensionless spin parameter a .

For the prograde ISCO, the classic Bardeen formula yields:

$$r_{\text{ISCO}}^{\text{pro}}(a) = 6M - \frac{4\sqrt{6}}{3}Ma + \mathcal{O}(a^2)$$

Here, $c_1 \approx -3.266 \neq 0$. The ISCO is **First-order Sensitive** to spin perturbations, as frame-dragging breaks the symmetric cancellation mechanism present in the static case.

Crucially, for the event horizon, the exact Kerr solution is $r_+ = M + \sqrt{M^2 - a^2}$. Expanding this for small a :

$$r_+(a) = 2M - \frac{a^2}{2M} + \mathcal{O}(a^4)$$

The first-order coefficient is strictly zero ($c_1 = 0$). The Kerr event horizon exhibits **Second-order Rigidity** (a^2 response) to spin perturbations, mirroring the structural rigidity observed in the Schwarzschild ISCO under metric deformations.

4.2 Photon Sphere Response (CH-S-0005)

For the equatorial prograde photon sphere, the exact Kerr solution is $r_{\text{ph}} = 2M \{1 + \cos[\frac{2}{3} \arccos(-a/M)]\}$. Perturbative expansion yields:

$$r_{\text{ph}}^{\text{pro}}(a) = 3M - \frac{2}{\sqrt{3}}a + \mathcal{O}(a^2)$$

Here, $c_1 = -2/\sqrt{3} \neq 0$. The photon sphere remains **First-order Sensitive**.

Summary of Kerr Hierarchy: Under spin perturbations, the hierarchy shifts. The photon sphere is First-order Sensitive (a^1), while both the ISCO and the Event Horizon exhibit Second-order Rigidity (a^2).

V. Generalized Perturbation Input Parser

CH-P-0001 establishes a generalized effective potential constructor

supporting Type B perturbations.

Type B Effective Potential (Joint g_{tt} and g_{rr} deformation):

$$V_{\text{eff}}^B = \frac{(1 - 2M/r)(1 + \alpha M/r)}{1 + \beta M/r} \left(1 + \frac{L^2}{r^2} \right)$$

Key Finding: The first-order perturbation of Type B depends solely on the difference $(\alpha - \beta)$. If $\alpha = \beta$ (i.e., symmetric deformations of equal magnitude to g_{tt} and g_{rr}), the first-order perturbation strictly vanishes. This provides a vital structural constraint: If a theory applies symmetric deformations of equal magnitude to both g_{tt} and g_{rr} , its ISCO structure will possess higher stability compared to when only g_{tt} is deformed.

VI. Broad Applicability and Implications

The core value of the CH Series lies in its broad applicability as a benchmark testing platform. Researchers of any modified gravity theory need only input the metric deformation form of their theory into the standard testing protocol of the CH Series to obtain the structural stability determinations.

Furthermore, the CH Series provides a powerful tool for verifying the internal self-consistency of specific modified gravity theories. By applying the model-independent CH protocol to a theory's numerical solutions, researchers can determine whether the observed structural shifts (e.g., significant ISCO deviations from the GR baseline of 6M) are merely first-order perturbative artifacts or genuine non-linear accumulations in the strong-field regime. This capability allows the CH Series to serve as an

independent cross-validation mechanism for complex numerical relativity simulations.

VII. Scope and Limitations of the Current Framework

The capability boundaries of the current CH Series framework are objectively defined as follows:

1. **Incomplete Photon Sphere Analysis for Kerr Backgrounds:** The current version only supports equatorial ISCO and photon sphere analysis on a Kerr background. Off-equatorial orbits require full three-dimensional effective potentials involving the Carter constant.
2. **Unverified Extremal Spin Limit:** The spin perturbation expansion has been numerically verified within the range $a \leq 0.3$. For extremal Kerr black holes ($a \rightarrow 1$), the ISCO behavior requires the full form of the Bardeen formula.
3. **Exclusion of Quasi-Normal Modes (QNMs):** Quasi-normal modes require an independent complex-frequency perturbation framework, which is not included in the current real-coordinate perturbation scheme.

VIII. Conclusion

We have established the CH Series framework for testing gravitational structural stability, examining the response behaviors of the event horizon, ISCO, and photon sphere under metric deformations and spin perturbations.

Our findings demonstrate that, under the specific Type-A metric deformation considered here, these three structures exhibit a distinct three-tier rigidity hierarchy on a Schwarzschild background. Most notably, the first-order rigidity of the ISCO is directly traced to the vanishing of the second derivative of the deformation function, $A_1''(6M) = 0$. Under spin perturbations on a Kerr background, the photon sphere exhibits an a^1 -order response, while both the ISCO and the event horizon exhibit a^2 -order rigidity.

This hierarchy, proven for the specific parametric deformations analyzed here, arises from distinct mathematical properties of the effective potential. The CH Series provides a systematic, mathematically rigorous toolchain for the structural analysis of gravitational theories, obviating the need for theory-specific derivations of stability conditions.

References

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Appendix A: Rigorous Proof of First-Order Rigidity for ISCO under Type-A Deformation

Objective: To prove that the first-order correction coefficient c_1 for the ISCO radius vanishes under the Type-A deformation $A(r) = \left(1 - \frac{2M}{r}\right) \left(1 + \alpha \frac{M}{r}\right)$.

Step 1: Formulation of the Characteristic Equation

For a static, spherically symmetric metric $ds^2 = -A(r)dt^2 + B(r)dr^2 + r^2d\Omega^2$, the effective potential for a test particle of unit mass is:

$$V_{\text{eff}}(r, L) = A(r) \left(1 + \frac{L^2}{r^2}\right)$$

The condition for a circular orbit is $V_{\text{eff}}'(r) = 0$, which yields the specific angular momentum squared:

$$L^2(r, \alpha) = \frac{r^3 A'(r)}{2A(r) - rA'(r)}$$

The ISCO is defined by the marginal stability condition $V_{\text{eff}}''(r) = 0$.

Substituting $L^2(r, \alpha)$ into $V_{\text{eff}}''(r)$ and clearing the denominator yields the characteristic equation for the ISCO radius:

$$F(r, \alpha) \equiv rA(r)A''(r) - 2r[A'(r)]^2 + 3A(r)A'(r) = 0$$

Step 2: Application of the Implicit Function Theorem

We seek the perturbed radius $r(\alpha) = 6M + c_1\alpha + \mathcal{O}(\alpha^2)$. Expanding

$F(r(\alpha), \alpha) = 0$ around $(r_0 = 6M, \alpha_0 = 0)$ gives:

$$c_1 = - \frac{\partial F / \partial \alpha}{\partial F / \partial r} \Big|_{r=6M, \alpha=0}$$

Step 3: Evaluation of Partial Derivatives at $(6M, 0)$

We decompose $A(r, \alpha) = A_0(r) + \alpha A_1(r)$, where:

$$A_0(r) = 1 - \frac{2M}{r}, \quad A_1(r) = \frac{M}{r} - \frac{2M^2}{r^2}$$

The partial derivative of F with respect to α at $\alpha = 0$ is:

$$\frac{\partial F}{\partial \alpha}|_{\alpha=0} = r(A_1 A_0'' + A_0 A_1'') - 4r A_0' A_1' + 3(A_1 A_0' + A_0 A_1')$$

We evaluate A_0, A_1 and their derivatives at $r = 6M$:

1. $A_0(6M) = \frac{2}{3}, A_0'(6M) = \frac{1}{18M}, A_0''(6M) = -\frac{1}{54M^2}$
2. $A_1(6M) = \frac{1}{9}, A_1'(6M) = -\frac{1}{108M}, A_1''(6M) = \frac{2M}{(6M)^3} - \frac{12M^2}{(6M)^4} = 0$

Substituting these values into the expression for $\frac{\partial F}{\partial \alpha}|_{\alpha=0}$:

- Term 1: $r A_1 A_0'' = (6M) \left(\frac{1}{9}\right) \left(-\frac{1}{54M^2}\right) = -\frac{1}{81M}$
- Term 2: $r A_0 A_1'' = (6M) \left(\frac{2}{3}\right) (0) = 0$
- Term 3: $-4r A_0' A_1' = -4(6M) \left(\frac{1}{18M}\right) \left(-\frac{1}{108M}\right) = \frac{1}{81M}$
- Term 4: $3 A_1 A_0' = 3 \left(\frac{1}{9}\right) \left(\frac{1}{18M}\right) = \frac{1}{54M}$
- Term 5: $3 A_0 A_1' = 3 \left(\frac{2}{3}\right) \left(-\frac{1}{108M}\right) = -\frac{1}{54M}$

Summing these terms yields the exact cancellation:

$$\frac{\partial F}{\partial \alpha}|_{r=6M, \alpha=0} = \left(-\frac{1}{81M} + \frac{1}{81M}\right) + \left(\frac{1}{54M} - \frac{1}{54M}\right) + 0 = 0$$

Step 4: Verification of the Non-Vanishing Denominator

$$\frac{\partial F}{\partial r}|_{r=6M, \alpha=0} = \frac{d}{dr} \left(\frac{2M}{r^2} - \frac{12M^2}{r^3} \right) \Big|_{6M} = -\frac{1}{54M^2} + \frac{1}{36M^2} = \frac{1}{108M^2} \neq 0$$

Conclusion: Since $\frac{\partial F}{\partial \alpha} = 0$ and $\frac{\partial F}{\partial r} \neq 0$, it rigorously follows that $c_1 = 0$. ■

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